APEX INTERNATIONAL ACADEMY

CBSE Senior Secondary Affiliation No. 1930482 | Academic Excellence & Research Division

OFFICIAL ACADEMIC SOLUTION & EVALUATION DOSSIER

VERIFIED & GRADED

ASSIGNMENT: Multiplication Tables Mastery 1-12

SUBJECT: Mathematics (MATH101)

CLASS & SECTION: Class 3 - Section A

ACADEMIC YEAR: 2025 - 2026

MARKS EVALUATED: 96 / 100 (Grade A)

TEACHER DIRECTIVE & PROBLEM FOCUS:
Practice and record speed drill sheet for tables 6 through 12 with commutative and distributive proofs.

1. Executive Overview & Problem Context

Multiplication represents iterative equal-group addition and foundational algebraic scaling. Mastery of single and double-digit multiplication matrices (1×1 through 12×12) provides the mental arithmetic fluency mandatory for long division, prime factorization, and fraction manipulation.

2. Core Analysis, Solutions & Systematic Working

- **Commutative Property Proofs (a × b = b × a):**

  Multiplication order does not alter the product. Demonstrated geometrically using rectangular dot arrays: a 7 × 9 array contains 63 units, identical in area to a 9 × 7 array rotated by 90 degrees.

- **Distributive Property Mental Shortcuts:**

  Decomposing challenging multipliers into friendly base-10 components: 8 × 12 = 8 × (10 + 2) = (8 × 10) + (8 × 2) = 80 + 16 = 96. 7 × 14 = 7 × (10 + 4) = 70 + 28 = 98. This eliminates rote memorization dependency.

- **Patterns in the 9s, 11s, and 12s Tables:**

  Table of 9 exhibits the constant digit sum property (9 × 4 = 36 -> 3 + 6 = 9; 9 × 7 = 63 -> 6 + 3 = 9). Table of 11 exhibits twin repeated digits up to 99. Table of 12 represents composite step counting (10x + 2x).

3. Detailed Data Matrix & Systematic Findings

Multiplier Pair	Repeated Addition	Distributive Breakdown	Product	Mastery Speed
7 × 8	8 + 8 + 8 + 8 + 8 + 8 + 8	(7 × 5) + (7 × 3) = 35 + 21	56	< 1.2 seconds
9 × 6	6 added 9 times	(10 × 6) - (1 × 6) = 60 - 6	54	< 1.0 seconds
12 × 7	7 added 12 times	(10 × 7) + (2 × 7) = 70 + 14	84	< 1.5 seconds
8 × 9	9 added 8 times	(8 × 10) - (8 × 1) = 80 - 8	72	< 1.1 seconds
12 × 12	12 added 12 times	(12 × 10) + (12 × 2) = 120 + 24	144	< 1.8 seconds

4. Student Conclusions & Reflective Synthesis

Instant recall of multiplication facts across tables 1 through 12 empowers young mathematicians to tackle multi-digit operations, estimation, and geometric area calculations with accuracy and speed.

5. Official Faculty Evaluation & Grading Rubric

ASSESSMENT CRITERIA BREAKDOWN

Conceptual Clarity & Factual Accuracy: 28 / 30

Analytical Depth & Research Methodology: 28 / 30

Problem Solving & Structured Working: 18 / 20

Presentation, Organization & Neatness: 14 / 20

FACULTY FEEDBACK REMARK:

"Outstanding analytical depth & speed! Commutative proofs and mental arithmetic decompositions are exemplary."

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